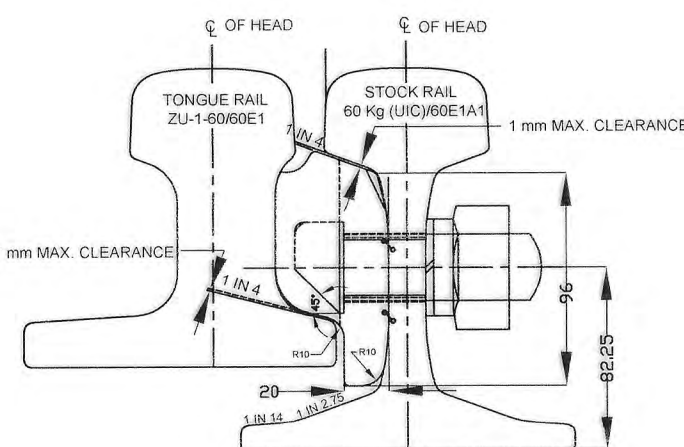
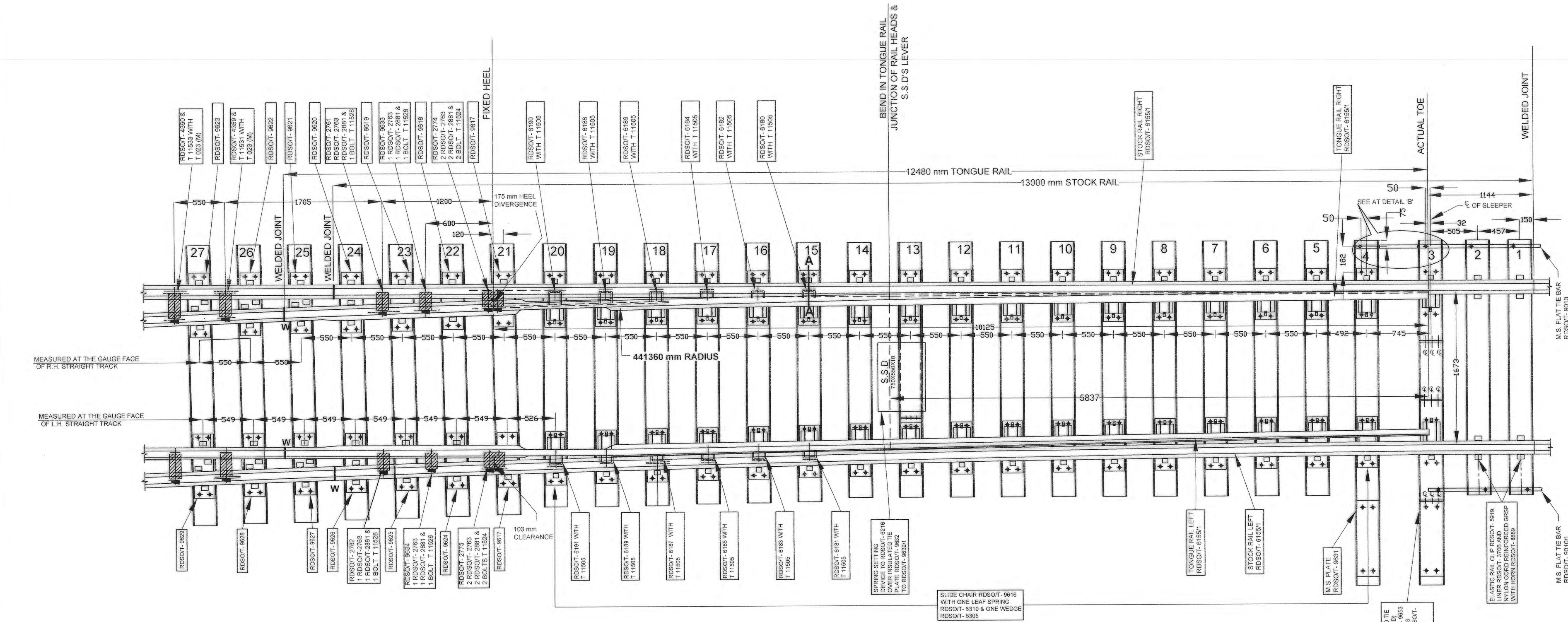
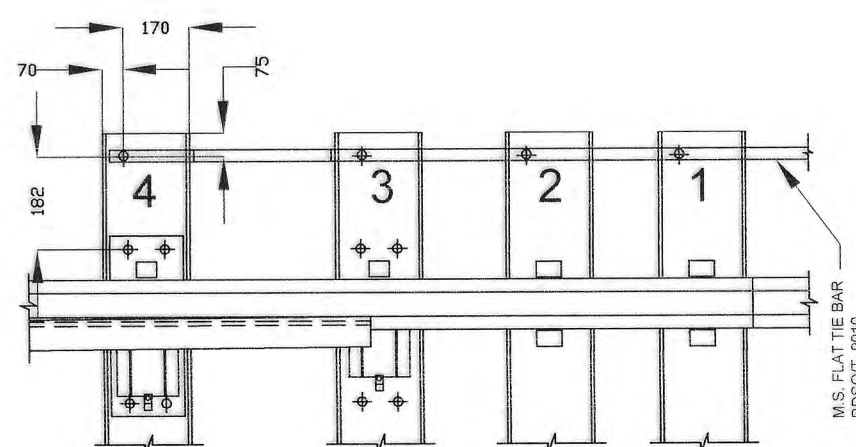
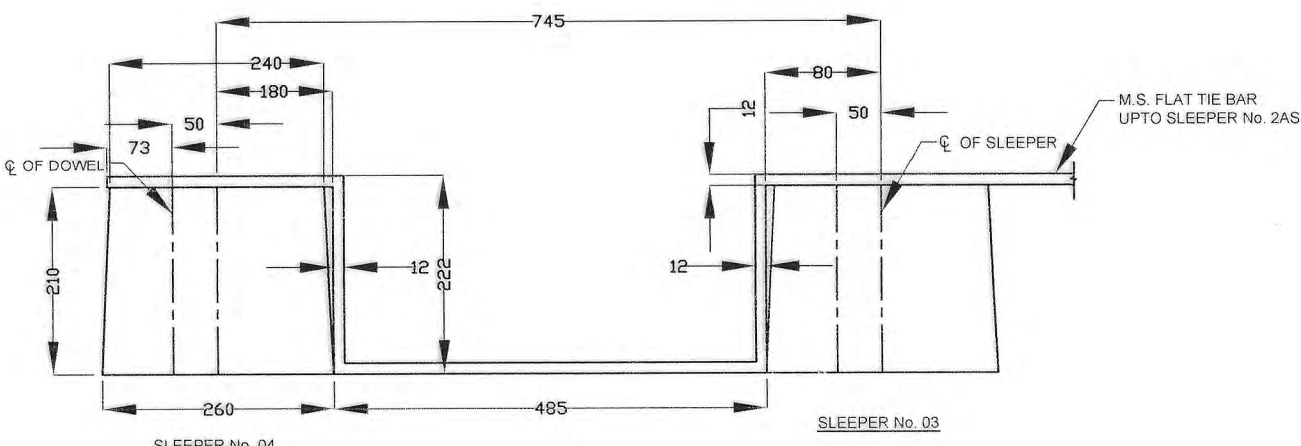




SECTION AT A-A



PLAN



ELEVATION

DETAIL 'B'
ARRANGEMENT OF M.S. FLAT TIE BAR AT SLEEPER No. 03 & 04

SWITCH ANGLE AT TOE-----0°- 20' - 0"

DIVERGENCE AT HEEL----- 175 mm

THROW AT TOE -----160 mm

GAUGE-----1673 mm

RADIUS-----441360 mm

TABLE OF SLEEPERS AND THEIR FITTINGS

S. NO. OF SLEEPER	SLEEPER TO DRG. NO.	LENGTH (mm)	INSULATED TILE PLATE ONE TO TWO WITH FITTINGS	OF NYLON CORD REINFORCED GRSP TO DRG. NO. RDSOT-	24 DIA. PLATE BOLTS RDSOT- 3913	INSULATING LINERS RDSOT- 3706	TWO SLOE CHAIRS	SINGLE COIL SPRING WASHER 11 0773	M/S. PLATE RDSOT- 6601	WEDGE TO DRG. NO. 6305	LEAF SPRING RDSOT- 6310	METAL LINERS RDSOT- 3740	ELASTIC RAIL CLIP MK-V RDSOT- 5919
1 & 2	RDSOT- 4512	2750	RDSOT- 9653 TO RDSOT- 9653G	8893 8854 8855 4 (8889)	2 EACH	4 EACH	-	2 EACH	-	-	-	-	- 4 EACH
3	RDSOT- 4514	2750	-	2 2 (8954)	2 (8955)	14	-	-	-	2 2	2 2	2 2	2 2
4	RDSOT- 4515	2750	-	2 2 (8955)	2 (8956)	13	-	RDSOT- 9616	-	1 2	2 2	2 2	2 2
5	RDSOT- 4516	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	-	-	2 2	2 2	2 2
6	RDSOT- 4517	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	-	-	2 2	2 2	2 2
7	RDSOT- 4518	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
8	RDSOT- 4519	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
9	RDSOT- 4520	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
10	RDSOT- 4521	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
11	RDSOT- 4522	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
12	RDSOT- 4523	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
13	RDSOT- 4524	2750	RDSOT- 9632 TO RDSOT- 9633T	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
14	RDSOT- 4525	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
15	RDSOT- 4526	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
16	RDSOT- 4527	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
17	RDSOT- 4528	2750	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
18	RDSOT- 4529	2770	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
19	RDSOT- 4530	2730	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2
20	RDSOT- 4531	2800	-	2 2 (8955)	-	8	-	RDSOT- 9616	8	-	2 2	2 2	2 2

S. No. OF SLEEPER	SLEEPER TO DRG. NO.	COL. LENGTH (mm)	SPECIAL PLAT BEARING PLATE TO DRG. NO.		24 DIA. PLAT SCREWS TO DRG. NO. RDSDOT- 3913	METAL LITERALS RDSDOT- 3740	ELASTIC ROLLER RDSDOT- 3740	SINGLE ROLLER RDSDOT- 3740	NYLON CORD REINFORCED GRG RDSDOT- 3740				
			RIGHT	LEFT					690T	8894	8895	8896	9630
21	RDSDOT- 4532	2820	RDSDOT- 9517	RDSDOT- 9517	8	2	2	8	-	2	-	2	-
22	RDSDOT- 4533	2830	RDSDOT- 9518	RDSDOT- 9524	8	4	4	8	-	2	2	-	-
23	RDSDOT- 4534	2840	RDSDOT- 9519	RDSDOT- 9525	8	4	4	8	-	2	2	-	-
24	RDSDOT- 4535	2870	RDSDOT- 9620	RDSDOT- 9603	8	6	6	8	2	-	2	2	2
25	RDSDOT- 4536	2890	RDSDOT- 9621	RDSDOT- 9527	8	8	8	8	-	2	-	2	4
26	RDSDOT- 4537	2900	RDSDOT- 9622	RDSDOT- 9629	8	8	8	8	-	2	-	2	4
27	RDSDOT- 4538	2920	RDSDOT- 9603	RDSDOT- 9629	8	8	8	8	-	2	-	2	4

LIST - A

T-11506	BOLTS	25X310		
RSOFT-5630	NYLON CON ROD REINFORCED GRIP		12	2
RSOFT-5635A	WEDGES			
RSOFT-5635	WEDGE		34	2
RSOFT-5957	NYLON CON REINFORCED GRIP			
RSOFT-5955	NYLON CON REINFORCED GRIP		42	2
RSOFT-5954	NYLON CON REINFORCED GRIP			
RSOFT-5980	NYLON CON REINFORCED GRIP		4	6
RSOFT-5986	NYLON CON REINFORCED GRIP			
RSOFT-5984	NYLON CON REINFORCED GRIP		6	2
RSOFT-5984	NYLON CON REINFORCED GRIP			
RSOFT-5993	NYLON CON REINFORCED GRIP			
RSOFT-5958	NYLON CON REINFORCED GRIP		36	4
RSOFT-6310	LEAF SPRING			
RSOFT-2681	PACKING PIECES			
RSOFT-2783	PACKING PIECES (TAPERED)		8	239
T-0377	PACKING ROLLING WASHERS			
RSOFT-3913	PLATE SCREWS		216	
T-11533	BOLTS	25X330	2	2
T-11531	BOLTS	25X326		
T-11529	BOLTS	25X326		
T-11524	BOLTS	25X292		
T-11505	BOLTS	25X108	4	12
T-023(M)	SPHERICAL WASHERS			

R-11526	BOLTS	25x310	1	2
RSDOT-9634	DISTANCE BLOCKS		1	2
RSDOT-9635	DISTANCE BLOCKS		1	2
RSDOT-9639	NYLON CORD REINFORCED GRSP		12	1
RSDOT-9650	ELASTIC RAIL CLIP MC/V		76	84
RSDOT-9740	METAL LINER		76	1
RSDOT-98101	M.S. FLAT THE BAR		1	1
RSDOT-9810	M.S. FLAT THE BAR		1	1
RSDOT-6217				
RSDOT-6217/39	DETAIL/COMPONENT OF S.S.D.		ONE SET	
RSDOT-6305A	WEDGE			
RSDOT-6305	WEDGE		34	2
RSDOT-8807	NYLON CORD REINFORCED GRSP		42	2
RSDOT-8885	NYLON CORD REINFORCED GRSP		4	2
RSDOT-8889	NYLON CORD REINFORCED GRSP		4	2
RSDOT-8896	NYLON CORD REINFORCED GRSP		6	2
RSDOT-8895	NYLON CORD REINFORCED GRSP		6	2
RSDOT-8894	NYLON CORD REINFORCED GRSP		36	4
RSDOT-8893	NYLON CORD REINFORCED GRSP		36	4
RSDOT-8895	NYLON CORD REINFORCED GRSP		36	4
RSDOT-6216	LEAF SPRING		1	2
RSDOT-6216	CHECK RAILS		8	8
RSDOT-6216	SPRING SETTING DEVICE		1	2
RSDOT-2581	PACKING		8	8
RSDOT-2763	PACKING PIERCES (TAPERED)		8	8
RSDOT-6631	M.S. PLATES			
RSDOT-3913	T. 203MM		239	215
RSDOT-3913	PLATE SCREWS			
T-11533	BOLTS	25x380	2	2
T-11531	BOLTS	25x360	2	2
T-11528	BOLTS	25x350	2	2
T-11524	BOLTS	25x290	2	2
T-11505	BOLTS	25x100	12	4
RSDOT-9617	CAST STEEL SPECIAL BEARING PLATES		1	2
RSDOT-9618	CAST STEEL SPECIAL BEARING PLATES		ONE EACH	
RSDOT-9629	CAST STEEL SPECIAL BEARING PLATES		ONE EACH	
RSDOT-9632	INSULATED THE PLATES FOR SLEEPER NO. 13		ONE SET	
RSDOT-9632/1	INSULATING LINER		6	
T- 023MM	SPHERICAL WASHERS		4	
RSDOT-9616	SLIDE PLATES		34	2
RSDOT-9653	CAST STEEL/MILD STEEL EXTENDED TIE PLATE (INSULATED) (ON S.L. No. 03)		ONE SET	
RSDOT-9653/3	METAL LINER		ONE SET	
RSDOT-6191	SLIDE BLOCKS		ONE EACH	
RSDOT-4380	DISTANCE BLOCKS		2	2
RSDOT-4359	DISTANCE BLOCKS		2	2
RSDOT-2762	DISTANCE BLOCKS		2	2
RSDOT-2761	DISTANCE BLOCKS		2	2
RSDOT-2774	HEEL BLOCKS		ONE EACH	
RSDOT-2775	HEEL BLOCKS		ONE EACH	
RSDOT-6155/1	TONGUE RAIL LEFT		1	1
RSDOT-6155/1	TONGUE RAIL RIGHT		1	1
RSDOT-6155/1	STOCK RAIL LEFT		1	1
RSDOT-6155/1	STOCK RAIL RIGHT		1	1
			No	

PART	DESCRIPTION	NO
THIS DRAWING IS THE PROPERTY OF RESEARCH DESIGNS AND STANDARDS ORGANISATION (MINISTRY OF RAILWAYS) LUCKNOW-226011, INDIA AND SHALL NOT BE USED, COPIED OR REPRODUCED IN PART OR WHOLE, WITHOUT PRIOR CONSENT IN WRITING.		
R.	D.	S. O.
10125 mm CURVED SWITCH WITH ZU-1-60/60E1A1 THICK-WEB TONGUE RAILS FOR 1 IN 12 TURNOUT B.G. (1673 mm) FOR 60 kg (IUC)/60E1 ON P.S.C. SLEEPERS		
STAMP _____ CANCEL _____ REV. _____	ADVANCE 28-11-2002	(T)
RDSO/T- 6155		

14. THE HEAD, WEAR AND FOOT OF EACH RAIL SHALL BE ULTRASONICALLY TESTED BY THE MANUFACTURER THROUGH A COMPETENT PERSON BEFORE USING THE RAILS FOR MANUFACTURE. A PERSON POSSESSING A COMPETENCY CERTIFICATE ISSUED BY DIRECTOR GENERAL (M & C) IRSD SHALL BE DEEMED A COMPETENT PERSON.
13. NYLON COIL REINFORCED GRSP SHALL BE USED UNDER SLIDE CHAIRS, TIE PLATES AND SPECIAL BEARING PLATES AS SHOWN IN THE TABLE ALONG SIDE.
11. SINGLE COIL SPRING WASHERS TO DRAWING NO. T 10773 SHALL BE USED WITH ALL PLATE SCREWS TO DRAWING NO. RDSOT- 3913.
11. THE PART LIST OF THIS DRAWING DOES NOT INCLUDE FITTINGS FOR APPROX SLEEPERS. THESE SHOULD BE INDENTED FOR SEPARATELY, IF REQUIRED.
10. HOLES FOR ADDITIONAL DISTANCE BLOCKS TO DRG. NO. RDSOT- 4359 & RDSOT- 4360 IN LEAD RAILS WILL NORMALLY BE DRILLED AT SITE.
9. SINGLE COIL SPRING WASHERS TO DRG. NO. T 10773 SHALL BE USED WITH ALL BOLTS OF 25 mm DIA.
8. THE HEADS OF BOLTS T 11524, T 11526 & T 11528 USED WITH BLOCKS SHALL BE LOCATED INSIDE THE TRACK GAUGE. THE WIDTH OF HEAD OF THESE BOLT SHALL BE REDUCED BY 4 mm BY MACHINING ON ONE SIDE TO ENABLE FIT.
7. P.S.C SLEEPERS USED IN THIS SWITCH ARE IDENTICAL TO THOSE USED IN 10125 mm CURVED SWITCH TO DRG. NO. RDSOT- 4219.
6. AT SLEEPER NO. 15, ONE SIDE 45° BEVEL CUT/HEAD BOLT T 11505 SHALL BE USED TO FASTEN SLIDE BLOCK WITH STOCK RAIL AS PER SECTION A-A.
5. THE STOCK RAILS ARE TO BE SUPPLIED BY THE MANUFACTURER. SLIDE BLOCKS SHALL BE BOLTED TO THE RESPECTIVE STOCK RAIL AND C.I. HEEL BLOCKS BOLTED IN PLACE AND THE TOE END OF THE TONGUE RAIL SHALL BE LASHED TO THE STOCK RAIL, WITH STOUT WIRE.
4. SLIDE CHAIRS AND BOLTS ARE SAME FOR BOTH SIDES.
3. THOSE BOLTS WHICH ARE SKEW TO THE RAILS SHALL HAVE SPHERICAL WASHERS T 023 (M) & TAPERED PACKING PIECES RDSOT- 2763 SHOWN ON THE DRAWING TONGUE - .
2. LATEST ALTERATION NUMBER SHALL BE CHECKED BEFORE PLACING ORDER.
1. ALL DIMENSIONS ARE IN MILLIMETRES.

20. 10% QUANTITY (i.e. 1/10th) NO. OF THE QUANTITY OF PURCHASE ORDER OF 1 IN 12, 60 kg (UCJ009E1 THICK WEB SWITCH) OF BREAKAGE PRONE AND WIRE PRONE SPARE PARTS (SHOWN IN LIST-1) DRG.NO. RDSDOT-01950 SHALL BE PROCURED WITH EVERY PURCHASE ORDER OF 1 IN 12, 60 kg (UCJ009E1) TMS FOR REPLACEMENT OF THESE SPARE PARTS AS PER NEED DURING SERVICE.
21. PRE-CURVING OF TONGUE RAILS & STOCK RAILS FOR TURNOUTS OF CONTRARY FLEXURE & SIMILAR FLEXURE SHALL BE DONE AT MANUFACTURER'S PREMISES AS PER THE CURVATURE OF MAIN LINE AND CORRESPONDING VEHICLE DETAILS GIVEN IN ANNEXURE - 4/6 & ANNEXURE - 4/7 OF IRPMV.
22. M.S. THE BAR TO DRG. NO. RDSDOT-0910 SHALL BE FIXED ON THE NON-POINT MACHINE END USING PLATE SCREW TO DRG. NO. RDSDOT-3913 AFTER 24 HOURS OF FIXING OF THE DOWEL.
23. IF DOWEL HOLES AT THE END OF SLEEPER NO. 3 & 4 ARE NOT AVAILABLE FOR FIXING OF M.S. THE BAR, ADDITIONAL HOLES OF 35 mm DIA. AND 165 mm DEPTH SHALL BE CAREFULLY DRILLED AT SHOWN IN DETAIL 'B' OF DRAWING NO. RDSDOT-01950 (AL1-12). POLYETHYLENE DOWEL TO DRG. NO. RDSDOT-3002 SHALL BE FIXED IN THE HOLE USING EPOXY RESIN OF TYPE L-100 CONFORMING TO IS-12994-1990.
24. M.S. THE BAR TO DRAWING NO. RDSDOT-90101 SHALL BE USED ON POINT MACHINE END AT SLEEPER NO. 2AS, 1AS, 1, 2, 3, 4 AS SHOWN ON LAYOUT DRAWING OF TURNOUT, RDSDOT-6154.
25. M.S. THE BAR TO DRAWING NO. RDSDOT-90101 SHALL BE USED ON NON-POINT MACHINE END AT SLEEPER NO. 2AS, 1AS, 1, 2, 3, 4, 4 AS SHOWN ON LAYOUT DRAWING OF TURNOUT.
26. IN CASE OF REWELDING OF JOINT AT SRJ, SLEEPER NO. 1 CAN BE SHIFTED TOWARDS INSIDE SLIGHTLY SO THAT THE WELDED JOINT REMAINS IN FLOATING CONDITION & HOLE IN MS PLAT THE BAR BE MADE ACCORDINGLY.
27. LEVER OF S.S.D. SHALL BE GIVEN IN 28 DIA. HOLES DRILLED IN FOOT OF BOTH TONGUE RAILS AT J.O.H. ALONG WITH INSULATING BUSH.
28. THE CURVED TONGUE & STOCK RAILS SHALL BE GIVEN THE CORRECT BEND AS SHOWN IN DRG. NO. RDSDOT-6154 BY THE MANUFACTURE BEFORE DESPATCH FROM HIS WORKS WHILE INDENTING FOR THE TURNOUTS, THE RAILWAY SHAL CLEARLY INDICATE THE NUMBER, L.H. & R.H. TURNOUT REQUIRED BY THEMAS THE CURVATURE OF TONGUE & STOCK RAILS GIVEN BY THE MANUFACTURER IS LIKELY TO GET DISTURBED DURING TRASIT TO THE SITE, SAME SHALL BE CHECKED & RECTIFIED BY THE S.E. (P.WAY) WHILE LAYING IN THE TRACK.
29. SLEEPER NO. 21 AND ONWARD ARE LAID IN FANPSE.
30. A SPRING SETTING DEVICE (S.S.D.) IS TO BE PROVIDED NEAR JUNCTION OF RAIL HEADS (J.O.H)
31. OPERATION OF SWITCH IS TO BE CARRIED OUT ELECTRICALLY NEAR TOE OF SWITCH.
32. CLAMP POINT LOCK TO DRG. NO. RDSDOS-3454 SHALL BE USED BETWEEN SLEEPER NO. 3 & 4 ALONG WITH 220 mm STROKE IRSP POINT MACHINE.
33. WHEN THIS ASSEMBLY IS USED WITH 1 IN 12 CMS CROSSING TO DRG. NO. RDSDOT-4220, THE CHECK RAIL SHALL ALSO BE PROCURED ALONG WITH COMPONENTS OF THIS SUB-ASSEMBLY.

NOTE

IRS: T 10.	NOT TO SCALE
SPECIFICATION	SCALE

13.	<u>REVISED & REDRAWN</u> (i) NOTE No. 28 ADDED (j) LIST - A DAAED	27-01-2025	AND SHALL NOT BE USED, COPIED OR REPRODUCED IN PART OR WHOLE, WITHOUT PRIOR CONSENT IN WRITING.
12.	<u>REVISED & REDRAWN</u> (a) NOTE No. 23, 24, 25, 26 & 27 ADDED. (j) BEND IN M.S. PLAT Tie BAR TO DRG. No. RDSQT- 9010 BETWEEN SLEEPER No. 63 & 64 MODIFIED	01-10-2024	R. D. S. O. 10125 mm CURVED SWITCH WITH ZU-1-60/60E1A1 THICK-WEB TONGUE RAILS FOR 1 IN 12 TURNOUT B.G. (1673 mm) FOR 60 kg (UIC)/60E1 ON P.S.C. SLEEPERS
11.	<u>REVISED & REDRAWN</u> (iii) PART LIST MODIFIED (i) NOTE No. 9, 11 &16 (ALT-10) DELETED (j) MACHINED JOINT 'M' OF TONGUE RAIL ARE REPLACED WITH WELDED JOINT 'W'	22-05-2024	STAND. CHARGE ADVANCE 28-11-2002 (T)
ALT:	DESCRIPTION	DATE	RDSO/T- 6155